

Hessian's Matrix

$$\hookrightarrow \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix}$$

f_x = Partial derivative wrt x

f_y = Partial derivative wrt y

f_{xx} = Partial derivative of f_x wrt x

f_{xy} = Partial derivative of f_x wrt y,

★ Then, the Hessian test can be applied to check the nature of the critical points.

1. $|Hf(a)| > 0$ and $f_{xx}(a) > 0$ then a is a local minimum.
2. $|Hf(a)| > 0$ and $f_{xx}(a) < 0$ then a is a local maximum.
3. $|Hf(a)| < 0$ then a is a saddle point.
4. $|Hf(a)| = 0$ then the test is inconclusive.

Q. The max. value of the function $-2x^2 + 2xy - y^2 + 12x - 4y - 7$

Sol.

$$f(x, y) = -2x^2 + 2xy - y^2 + 12x - 4y - 7$$

$$R_x \Rightarrow -4x + 2y + 12 = 0$$

$$R_y \Rightarrow 2x - 2y - 4 = 0$$

$$\begin{matrix} \nearrow \\ \searrow \end{matrix} \begin{matrix} x = 4 \\ y = 2 \end{matrix} \rightarrow \text{Critical point}$$

$$\text{Hessian: } \begin{bmatrix} -4 & 2 \\ 2 & -2 \end{bmatrix}$$

$$\begin{matrix} \hookrightarrow \\ \hookrightarrow \end{matrix} \begin{matrix} D = 4 > 0 \\ f_{xx} = -4 < 0 \end{matrix} \rightarrow \text{Maximum}$$

$$f(4, 2) = 13 \rightarrow \text{max}$$

Q. A particle moving in a potential field tends to settle at points where the potential energy is a local minimum (stable equilibrium point). Consider a particle moving in $\mathbb{R}^2$ where the PE as func. of position is given by $V(x, y) = x^4 + y^4 - 4xy + 1$ for all $(x, y) \in \mathbb{R}^2$. Using the Hessian test, Find the PE at stable equilibrium points.

Sol.

$$V(x, y) = x^4 + y^4 - 4xy + 1$$

$$\begin{aligned} V_x &= 4x^3 - 4y = 0 \Rightarrow y = x^3 \\ V_y &= 4y^3 - 4x = 0 \Rightarrow x = y^3 \end{aligned} \left\} \begin{array}{l} \rightarrow \text{Solve system} \\ x = 0, \pm 1 \end{array} \right.$$

Hessian:

$$V_{xx} = 12x^2, \quad V_{yy} = 12y^2, \quad V_{xy} = -4$$

$$H = \begin{bmatrix} 12x^2 & -4 \\ -4 & 12y^2 \end{bmatrix}$$

Check each point:

$$\text{at } (0, 0) \rightarrow H = \begin{bmatrix} 0 & -4 \\ -4 & 0 \end{bmatrix} \xrightarrow{|D|} -16 < 0 \rightarrow \text{saddle}$$

$$\text{at } (1, 1) : \rightarrow H = \begin{bmatrix} 12 & -4 \\ -4 & 12 \end{bmatrix} \xrightarrow{|D|} 128 > 0$$

~~at $(-1, -1)$~~

at $(-1, -1) \rightarrow \text{also min}$

$V_{xx} > 0$
(minimum)

At $(1, 1)$ and $(-1, -1)$ can give $\boxed{-1}$ Ane.
PE at stable equi. pos.